

Project Plan Internship assignment

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1. Introduction

In this document, you will find the steps that will be taken to realize my internship assignment. First, you will get a general description of the assignment. After which you will be up to speed on what the internship assignment entails. Followed by a feasibility study done on various geocoding tools, also a comparison study is done on various libraries to be used in the assignment. Finally, you will be met with my project plan. This plan will explain in detail how all the requirements of the assignment will be met.

2. Tokheim

The internship will take place in the company Tokheim. They are a worldwide manufacturer in the petroleum industry. They provide a variety of products ranging from fuel dispensing equipment to payment terminal technology. They have a long-standing history in the market with the founder John J. Tokheim. He started the company in 1901 and patented the world's first vehicle fuel dispenser. Tokheim specializes in providing "Station-in-a-Box" solutions, allowing retailers to source nearly all their equipment from a single provider.

Since 2016, Tokheim operates as a core brand within **Dover Fueling Solutions (DFS)**, a division of



the [Dover Corporation](#), side by side to other major brands like Wayne Fueling Systems

Some of their products:

- **Fuel Dispensers:** The flagship **Quantum series** multiple fuel types.
- **Retail Automation:** Systems like the **Prizma POS** and **DFS Anthem UX** platform manage sales, inventory, and customer engagement.
- **Payment Solutions:** Secure terminals such as **Crypto VGA** and **Tokheim OASE** for multiple payment methods, including contactless transactions.
- **Clean Energy:** Expansion into alternative fuels including **AdBlue**, **CNG**, **LNG**, **Hydrogen**, and **EV charging** infrastructure.

Market Position & Future

As part of DFS, which reported approximately **\$7.7 billion in revenue** for 2024, Tokheim is a leading player in a market valued at roughly \$2.82 billion globally. The company is currently pivoting toward digital transformation and sustainability, investing in **IoT-enabled remote monitoring** and infrastructure for the energy transition away from fossil fuels.

(Matrix, 2025)

3. Assignment description

3.1. Introduction

In the following part, a detailed description of the internship assignment can be found. Starting with some context surrounding the work environment and tooling. Then the assignment description is put to light. Followed by what problem this solution will ultimately solve. We will also go into who is going to be using the end product. After this section, you will have a detailed sense of what the internship assignment entails and the goals this assignment aims to achieve.

3.2. Context

ONE PORTAL

In the current state, clients can use One Portal, an online management tool developed in Angular typescript, to manage their stations. Inside this tool, they have access to many different modules. Just to sum up a few:

- The stock module to manage their stock supply
- The pricing module to set the prices of the products in store
- The sales module to get statistics and insights into their sales

The alerts module is one of these. This module provides them with detailed messages and warnings about the status of their stations. For example, when a station's OPT (Outdoor Payment Terminal) is out of service. Then a red alert warning will show up in the One Portal alerts module. Or if a fuel tank is almost empty, an automatic event is sent and automatically acted upon, in order to schedule a delivery.

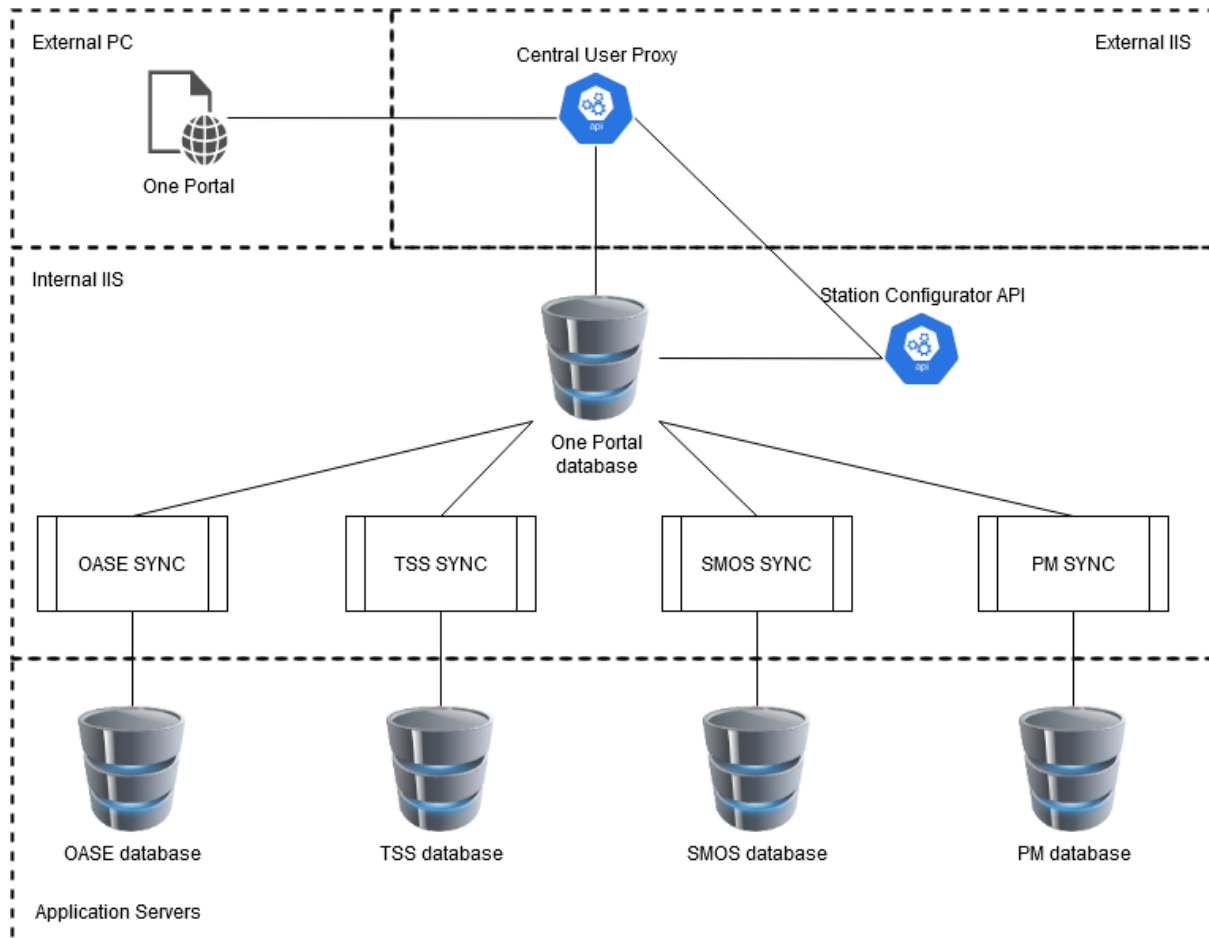
Right now, the alerts module is displayed in a tree view, where you can drill down on station networks to get the station alert details of a station. Admin users see all stations displayed in the view, but customers see only the stations of their own. This means the stations are loaded based on the group the client is connected to in the database.

The main goal of this module is to help clients keep an eye on their stations, help them spot issues fast and act quickly on them.

PETROL MANAGER

Petrol Manager is the old tool that was replaced by One Portal. Most of the functionalities of One Portal, stem from Petrol Manager.

DATABASE



1 CENTRAL DATABASE MECHANISM + PM SYNC

The database is developed in SQL-Lite and uses a synchronization mechanism to keep the central One Portal Database up to date. This simplifies the .NET API's that need to be written in order to retrieve the data. (D.Holemans, 2020)

The scope of the assignment is set to the PM (Petrol Manager) database, which contains the PetrolManagerConfiguration database and PetrolManagerEvent database. The first one contains information about the station's address. The latter contains all tables relevant to the alerts module.

3.3. Assignment Description

The internship assignment revolves around adding a new functionality to the alerts module. Currently the client can see his stations in a tree view grouped in station groups, just like displayed on the right. The functionality that clients have requested is to provide an interactive map containing all their stations. Important note here is that the users have certain permissions and can only see their stations. Other station networks are already filtered out by a custom filtering system.

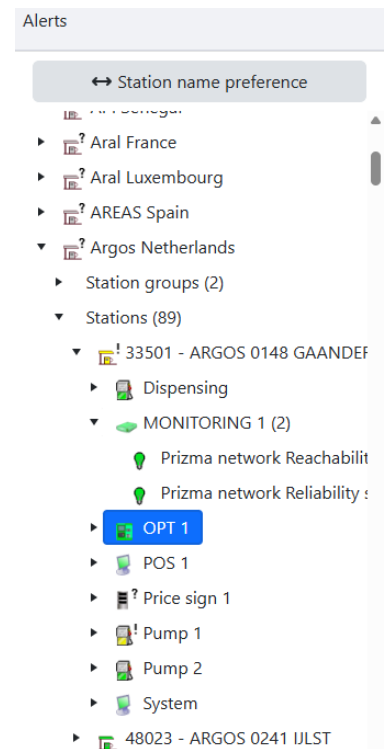
On this map, they will be able to perform the same actions as on the tree view. Only with this new view, they have a nice visual representation of their station network at all times. The map is tied into the alerts module, so that the information from the alerts is also able to be read from the markers on the map. The markers can display the severity or alert level of the message too using color codes. This means, requests will have to be made towards the API endpoints from Petrol Manager (PM is the old management platform before One Portal), to provide the right data to the map.

Finally, there is also the possibility to drill down on the map markers, to redirect to the alert group details page. If this redirect is not possible, the alert group details are instead displayed next to the map tick in a user-friendly format.

In preparation, a comparison study is also expected to be done on possible libraries to be used for the interactive map. This study is also a part of the internship assignment and will determine which library will be used to build the map.

Lastly, a feasibility study is done on ways to convert the addresses of the stations into coordinates. For this a tool will need to be made to convert the existing stations efficiently and accurately. A secondary assignment to this is to look into a way to automatically provide a newly created station with coordinates upon creation in the station configurator.

An important point is that this assignment was made without clear boundaries by design. The finished product of this assignment is still fluid and a lot of decisions on how to get to the end product, are intentionally left to me. This freedom gives me a lot of autonomy in how I can work out this assignment, but at the same time, a clear finish line is not defined. There are a lot of extra functionalities that can be added to the map. It is up to me to communicate clearly and on time together with my mentor, on what functionalities will be added to the map. Because of the nature of the assignment, a lot of details on what the map will look like and what it will be capable of, shall be left unsaid and left under the term “building a map”. The basic functionalities are described, but undoubtedly there will be extensions in the process if realizing the assignment.



2 Current OP Tree View

3.4. Goals

The end goal of this assignment is to implement the interactive map in One Portal, so that clients can put up a monitor in their station or control center with the interactive map on display. Furthermore, enabling them to monitor their station network continuously and effectively.

A secondary goal is to have a system in place to provide newly created stations with the right coordinates in the station configurator.

This means that the focus of the map should lie on ease of use and user-friendliness, as it is meant to be an alternative to the current tree view. The accent of the tree should remain on easy navigability and the accent of the map on having a high-level overview. The scenario we want to avoid is that the map becomes an identical copy of the tree view. In that case, the map would not provide any new value.

3.5. Conclusion

At the end of the internship, the following items will have been realized.

1. A feasibility study on conversion methods and tools of station addresses to coordinates.
2. A study which compares different libraries to be used for the interactive map.
3. The implementation of the interactive map in One Portal. Filling up the map with the right data from the API endpoints from Petrol Manager.
4. Providing extra functionality with the map. Configuring the alerts, providing a way to drill-down on a map marker and filtering on the right alert groups based on the permissions of the user.
5. An extension to the station configurator to provide the coordinates to new stations

4. Planning and deliverables

The internship starts at 22 February and end 22 May. In the first week, I will get a tutorial assignment to acclimatize to the new working environment and the code base.

Every week, my mentor has a small one on one session planned in to check in.

At the end of the internship, I will have the opportunity to present the result in front of the company.

Deadlines:	
Week 2: Friday 6 March	Kickoff meeting
Week 4: Thursday 18 March	Check in meeting
Week 7: Thursday 9 April	Intermediate evaluation
Week 9: 20-24 April	Second check in meeting
Week 12: Sunday 17 May	Submit documents for review
Wednesday 27 May	Jury practise sessions
Wednesday 17 June or Friday 19 June	Jury moment

5. References

D.Holemans. (2020, September 28). Feasibility Study One Portal Database.

Matrix, B. (2025, December 4). *How Does Tokheim S.A.S. Company Work?* Opgehaald van BCG Matrix:

Tokheim. (sd). *Tokheim*. Opgehaald van Tokheim: <https://www.tsg-solutions.com/be/nl/tokheim/>